



SignSpeaker — Phase 2 Engineering Plan

Integration of 5 Flex Sensors with Gyroscope Fusion & Full Pipeline Architecture

VERSION 2.0 (PLAN)
STA x AIO ROADMAP
SEPTEMBER 2026

Document Scope & Executive Context

This engineering plan specifies the full hardware, firmware, Node-RED backend, and Dashboard machine learning evolution required to expand the SignSpeaker smart glove from Phase 1 (wrist rotation via MPU6050 gyroscope only) to Phase 2 (5-finger bend articulation fused with 3-axis gyroscope kinematics). **Scope: Architectural plan and technical design only — no modifications are made to active Phase 1 production code.**

1. The Problem Statement & Physical Principles

Phase 1 Capabilities (Gyroscope Only)

The single MPU6050 IMU captures wrist and forearm rotational angular velocities (g_x , g_y , g_z) at 25 Hz. It excels at measuring macroscopic gesture dynamics:

- Directional arm sweeps (left-to-right, up/down arcs)
- Wrist rolls, flicks, and rotation velocity profiles
- Macroscopic spatial energy triggers for gesture boundaries

Phase 1 Fundamental Blind Spots

The gyroscope cannot measure hand topology or internal articulation:

- Finger curl:** Open palm vs. clenched fist produce 0 rad/s when static.
- Static signs:** Letters "A", "B", "C", "D" have identical zero gyro movement.
- Sign ambiguity:** "Hello" (open wave) and "Thanks" (flat palm push) share similar wrist arcs but have completely distinct finger shapes.

Egyptian Sign (ESL)	Phase 1 Gyroscope Ambiguity	Phase 2 Flex Sensor Fusion Resolution
"Hello" vs. "Thanks"	Both execute a forward/lateral wrist movement; angular momentum overlaps significantly.	"Hello" shows oscillating slight curl across all fingers; "Thanks" features all 5 fingers fully flattened ($f_0..f_4 < 0.10$) throughout the push.
"I am" vs. "Hello Reem"	Both involve downward wrist deflection towards chest/waist.	"I am" isolates index finger ($f_1 < 0.10$) with thumb/middle/ring/pinky curled tight ($f_0, f_2, f_3, f_4 > 0.85$); "Hello Reem" presents a distinct open-hand pose.
Alphabet & Digits	Completely unrecognizable if held stationary (gyro outputs near-zero noise).	Static poses are uniquely classified by finger bend configuration vectors alone (100% discriminatory flex features).

2. Hardware Architecture & Pin Allocation

2.1 Flex Sensor Principle & Voltage Divider

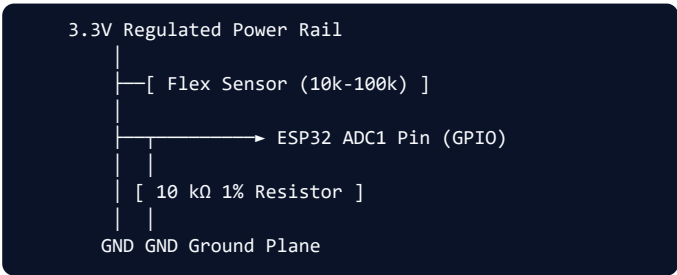
Flex sensors are carbon-based variable resistors whose resistance increases monotonically with bend angle:

- Flat (0° extension):** Nominal resistance $\approx 10\text{--}25\text{ k}\Omega$
- Curled (90°+ flexion):** Resistance increases to $60\text{--}110\text{ k}\Omega$

A passive voltage divider with a fixed $10\text{ k}\Omega$ pull-down resistor converts resistance changes to a voltage read by the ESP32 12-bit ADC (0–4095 counts):

$$V_{out} = 3.3V \times R_{fixed} / (R_{fixed} + R_{flex})$$
$$ADC_{Raw} = (V_{out} / 3.3V) \times 4095$$

2.2 Circuit Diagram per Finger



⚠ Critical Hardware Constraint: ADC1 Channels Only

The ESP32 microcontroller features two ADC units. **ADC2 is multiplexed with the internal Wi-Fi peripheral**; invoking Wi-Fi operations (MQTT publishing) immediately corrupts ADC2 readings. Therefore, **all 5 flex sensors must be mapped strictly to ADC1 channels (GPIO 32–39)** to guarantee zero radio interference. Note that GPIO 34, 35, 36, and 39 are input-only pins with no internal pull-ups, perfectly suited for external divider circuits.

Anatomical Finger	Mounting Location	Assigned ESP32 Pin	ADC Channel	Electrical Characteristics & Role
Thumb	Dorsal proximal phalanx	GPIO 34	ADC1_CH6	Input-only; 12-bit SAR ADC; external 10k reference
Index	Dorsal knuckle to PIP joint	GPIO 35	ADC1_CH7	Input-only; high dynamic range for pointing gestures
Middle	Dorsal knuckle to PIP joint	GPIO 32	ADC1_CH4	ADC1 analog input; general articulation tracking
Ring	Dorsal knuckle to PIP joint	GPIO 33	ADC1_CH5	ADC1 analog input; finger grouping discrimination
Pinky	Dorsal knuckle to PIP joint	GPIO 36 (Sensor_VP)	ADC1_CH0	Input-only; low-noise channel for small finger curl

SignSpeaker — Phase 2 Extended Protocol & Firmware

Telemetry Packet Design, On-Device Normalization, and ESP32 Pipeline

SECTION 3 & 4

3. Extended Telemetry & Packet Protocol

To support both Phase 1 (gyro-only) and Phase 2 (gyro + flex) hardware transparently without protocol breakages, telemetry strings transition from a 5-field structure to a 9-field standardized CSV line.

Phase 1 Legacy Packet (5 Fields)

```
D,<ms>,<gx>,<gy>,<gz>
```

Carries timestamp and 3 floating-point angular velocity axes.

Phase 2 Extended Packet (9 Fields)

```
D,<ms>,<gx>,<gy>,<gz>,<f0>,<f1>,<f2>,<f3>,<f4>
```

Carries timestamp, 3 gyro axes, and 5 normalized finger bend values.

Index	Field Token	Data Type & Range	Description & Engineering Semantics
0	D	Header String	Packet type discriminator. Identifies telemetry streaming frame.
1	<ms>	Unsigned Long	ESP32 internal timer <code>millis()</code> elapsed since boot.
2..4	<gx>,<gy>,<gz>	Float (rad/s)	MPU6050 angular velocity axes calibrated with moving EWMA zero-bias.
5	<f0>	Float (0.00 – 1.00)	Thumb bend index (0.00 = completely extended flat, 1.00 = fully bent).
6	<f1>	Float (0.00 – 1.00)	Index finger bend index.
7	<f2>	Float (0.00 – 1.00)	Middle finger bend index.
8	<f3>	Float (0.00 – 1.00)	Ring finger bend index.
9	<f4>	Float (0.00 – 1.00)	Pinky finger bend index.

On-Device Normalization & Flash Storage

To prevent floating-point transmission overhead and avoid raw ADC calibration drifts across ambient temperatures, normalization executes on the ESP32 transmitter:

```
f_norm[i] = clamp( (ADC_raw[i] - ADC_flat[i]) / (ADC_bent[i] - ADC_flat[i]), 0.0f, 1.0f )
```

Calibration pairs (`ADC_flat[5]` , `ADC_bent[5]`) are stored permanently in ESP32 non-volatile storage (NVS via `Preferences.h`), persisting across reboots.

ESP-NOW Wireless Struct Evolution

```
// Phase 1 Struct (12 bytes)
typedef struct {
    float gyroX;
    float gyroY;
    float gyroZ;
} GyroData;
```

```
// Phase 2 Struct (33 bytes)
typedef struct {
    float gyroX, gyroY, gyroZ; // 12 bytes
    float flex[5];             // 20 bytes
    uint8_t version;           // 1 byte (0x02)
} SensorData;
```

4. Firmware Architecture Evolution

1. Multi-Sensor Loop (25 Hz)

MPU6050 sampled over I2C (400 kHz); 5 ADC channels sampled via `analogRead()` .

2. Anti-Noise EWMA Filtering

ADC jitter dampened with exponential filter ($\alpha = 0.15$) on each finger channel.

3. Normalization & Clipping

Raw readings converted to 0.0–1.0 bounds via calibrated NVS constants.

4. Transmit (ESP-NOW / MQTT)

Packed into 33-byte struct or published as 9-field CSV over Wi-Fi.

4.1 Sender Firmware (`sender_v2.ino`)

- Maintains exact 40 ms cycle time (25 Hz) for fixed temporal resolution.

4.2 MQTT Sender (`mqtt_sender_v2.ino`)

- Increases PubSubClient buffer from 256 to **512 bytes** to prevent truncation.
- Publishes 9-field CSV string to `signspeaker/glove/data` .

- Implements interactive calibration triggered by button long-press or serial command `C`.
- Smooths sensor readings with EWMA filter: $y[n] = \alpha \cdot x[n] + (1 - \alpha) \cdot y[n-1]$.
- Keeps ADC1 isolated from Wi-Fi modem transients using local bulk decoupling capacitor.

💡 Two-Step Guided Calibration Routine

Step 1 (3 seconds): User holds hand completely flat on a table. Firmware averages 75 samples per finger to determine `ADC_flat[0..4]`.

Step 2 (3 seconds): User curls fingers firmly into a fist. Firmware averages 75 samples to record `ADC_bent[0..4]`. Parameters commit to flash and confirm via `I,CALIB_OK`.

SignSpeaker — Node-RED Pipeline & ML Feature Engineering

Connection Manager, Segmentation Engine, and 121-Dimensional Vector Architecture

5. Node-RED Processing Pipeline Architecture

5.1 Connection Manager (serial-manager.js)

The parser dynamically checks incoming token counts:

- **5 tokens:** Emits legacy payload {t, dev, gx, gy, gz, v:1} .
- **9 tokens:** Emits Phase 2 payload {t, dev, gx, gy, gz, flex:[f0..f4], v:2} .

The internal simulator is upgraded to synthesize realistic 5-channel finger trajectories alongside gyro waveforms, enabling testing of the entire stack without hardware.

5.2 Gesture Segmentation Engine (gesture-engine.js)

Crucial Architectural Decision: Motion triggers remain driven strictly by **gyroscope magnitude**:

$$\text{mag} = \sqrt{gx^2 + gy^2 + gz^2}$$

Flex sensors are not motion onset triggers — they define topological configuration during motion. The pre-roll buffer and state machine carry 9-element sample rows [t, gx, gy, gz, f0, f1, f2, f3, f4] seamlessly.

5.3 Feature Vector Architecture: Expansion from 66D to 121D

To empower the k-NN classifier to distinguish subtle finger configurations without losing temporal motion trajectory data, the feature vector expands from 66 to **121 dimensions** by appending a dedicated 55-dimensional flex block.

Feature Indices	Sub-Block Name	Dims	Mathematical Formulation & Engineering Purpose
[0]	Duration	1D	Total gesture elapsed time in milliseconds: $t_{end} - t_{start}$.
[1..15]	Gyro Per-Axis Stats	15D	5 statistical moments (Range, Mean, StdDev, Mean Absolute Value, Peak) across Gyro X, Y, Z.
[16..17]	Gyro Magnitude	2D	Mean energy and maximum peak of angular velocity magnitude vector.
[18..65]	Gyro Trajectory	48D	Temporal waveform resampled to 16 equidistant time-slices × 3 axes (X, Y, Z).
— NEW PHASE 2 FLEX FEATURE BLOCK (55 DIMENSIONS) —			
[66..90]	Flex Statistical Distribution	25D	5 statistical metrics per finger (Thumb, Index, Middle, Ring, Pinky): • Range: $\max(f) - \min(f)$ (measures finger dynamic activity) • Mean: $\mu(f)$ (average bend level during the gesture) • StdDev: $\sigma(f)$ (bend variance/stability) • Mean Absolute Value (MAV): Overall finger engagement energy • Absolute Peak: Extreme flexion point reached during the sign
[91..95]	Mean Static Pose	5D	Time-averaged bend of each finger across the entire gesture. Acts as an invariant "hand posture fingerprint" for static signs.
[96..105]	Onset Posture Snapshot	10D	Finger posture at the exact instant the gyro motion trigger tripped (first 2 time-slices × 5 fingers). Discriminates signs that start with identical hand shapes before diverging.
[106..120]	Coarse Flex Trajectory	15D	Temporal flex evolution resampled to 3 equidistant stages (Start, Midpoint, Finish) × 5 fingers. Captures opening/closing gestures without bloating dimensionality.
TOTAL DIMENSIONALITY		121D	66D Gyro Kinematics + 55D Flex Topology = Complete Gesture Representation

⚠ Dataset Compatibility & Training Segregation Rule

Phase 1 and Phase 2 training recordings must never be mixed within the same trained model. Zero-padding missing flex features on legacy Phase 1 recordings introduces severe artificial geometric bias into the 121D Euclidean distance space. When Phase 2 hardware is deployed, take an archive snapshot of Phase 1 model/dataset, clear the active training set, and collect clean Phase 2 samples.

Model Serialization Schema

```
{
  "version": 2,
  "featureCount": 121,
  "channels": [ "gx", "gy", "gz", "f0", "f1", "f2", "f3", "f4" ],
  "flexEnabled": true,
```

Distance Metric Formulation

Standard k-NN calculates Euclidean distance over Z-score standardized features. Optionally, Phase 2 supports a weighted distance metric balancing gyro and flex sub-spaces:

```
"k": 5,  
"labels": ["Hello", "I am", "Reem", "Thanks", ...],  
"samplesCount": 140  
}
```

$$D = \sqrt{W_{\text{gyro}} \cdot \Sigma(\Delta Z_{\text{gyro}})^2 + W_{\text{flex}} \cdot \Sigma(\Delta Z_{\text{flex}})^2}$$

Default: $W_{\text{gyro}} = 1.0$, $W_{\text{flex}} = 1.2$ to emphasize static finger discrimination.

SignSpeaker — Dashboard Evolution & Machine Learning

Live Articulation Visualizers, Calibration Wizard, and Classification Benchmarks

SECTION 6 & 7

6. Dashboard Interface Enhancements

6.1 Live Sensor Panel: 5-Finger Articulation Bars

Below the 60 FPS Gyroscope Canvas chart, a dedicated **Flex Telemetry Gauge** visualizes finger posture in real time (25 Hz):

- 5 dynamic vertical bar meters with color-coding:
 - Thumb (T):** Amber (#f59e0b)
 - Index (I):** Cyan (#06b6d4)
 - Middle (M):** Emerald (#10b981)
 - Ring (R):** Purple (#8b5cf6)
 - Pinky (P):** Orange (#f97316)
- Fills proportionally from 0.00 (flat) to 1.00 (flexed). Displays numeric readouts and highlights disconnected sensors.

6.2 Calibration Wizard Modal

An interactive modal accessible from the header and Training Studio guides users through glove calibration:

- "Open Palm Flat":** Visual hand illustration with 3-second progress countdown bar.
- "Clench Tight Fist":** Visual fist illustration with 3-second countdown bar.
- Verification Table:** Displays captured flat/bent ADC delta per finger. Flags weak sensors (delta < 800 counts).

6.3 Last Gesture Trace: Finger Timeline

The captured gesture card expands to include a 5-row temporal bend chart beneath the 3-axis gyro waveform, illustrating how fingers flexed or released throughout the active gesture window.

6.4 Feature Inspector: Grouped 121D Vector

The inspection modal breaks the 121D feature vector into 3 distinct color-coded bar chart groups: *Gyro Kinematics* (0..65), *Flex Distribution* (66..105), and *Flex Trajectory* (106..120).

7. Machine Learning & Classification Analysis

Projected Accuracy & Vocabulary Expansion

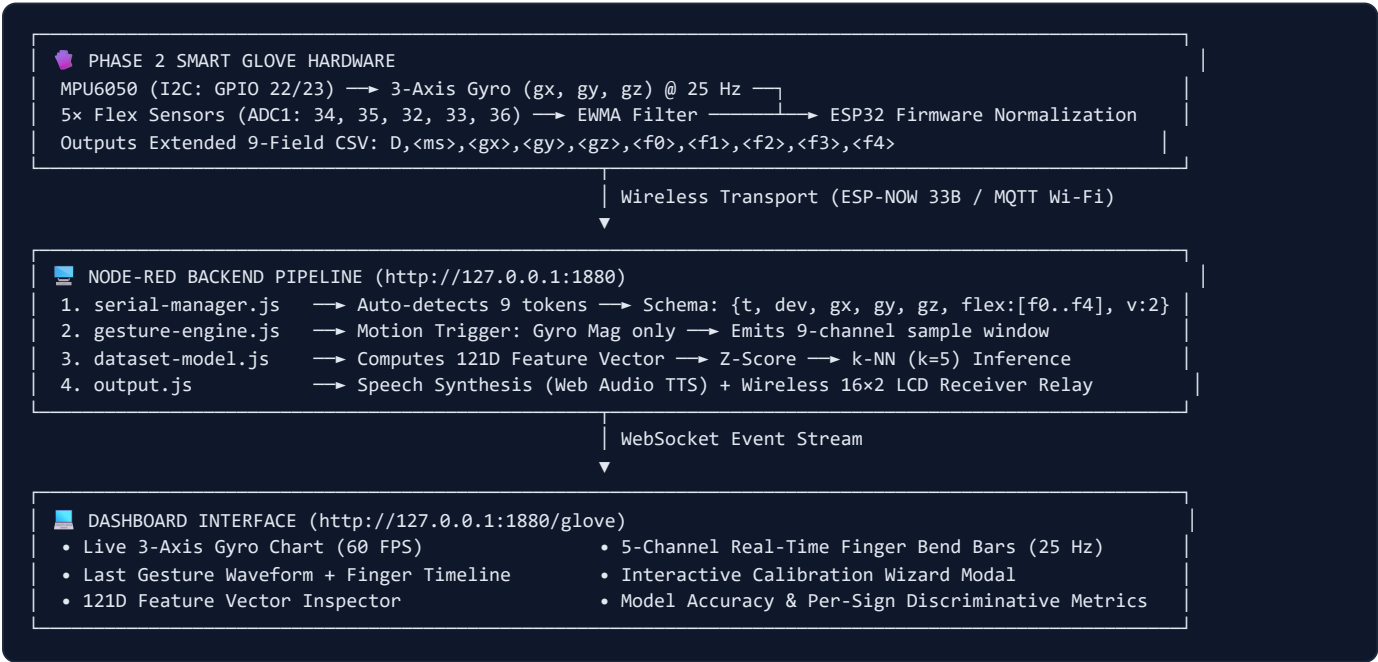
Phase 1 achieves 75–85% accuracy on 7 dynamic gestures, but fails on static signs. Phase 2 models are projected to reach **92–97% accuracy** across extended vocabulary:

- Completely eliminates confusion between outward waves and flat pushes.
- Enables single-handed alphabet signs (ESL / ASL fingerspelling).
- Supports complex two-phase compound words.

Sample Requirements & Hyperparameters

- Sample Density:** Increases from 3–5 to **10–20 samples per sign** to populate the 121-dimensional manifold adequately.
- k-NN Neighbors:** k parameter increases from `k = 3` to `k = 5` to average across higher-dimensional variance.
- Inference Latency:** Computation remains under **4.5 ms** in Node-RED, easily satisfying real-time 40 ms cycle budgets.

8. End-to-End System Pipeline Flow



SignSpeaker — Roadmap, Risk Matrix & Architectural Decisions

Implementation Checklist, Failure Mode Mitigations, and Final Engineering Summary

SECTION 9, 10, 11, 12

9. Simulator Extension Matrix for Phase 2 Testing

Sign Token	Simulated Gyroscope Kinematics	Simulated 5-Channel Flex Patterns	Verification Objective
Hello	Sinusoidal X/Z oscillation (wave)	All fingers extended (f0..f4 ≈ 0.10) with mild oscillation	Verifies dynamic hand + open palm fusion
I am	Downward vertical Y deflection	Index flat (f1 ≈ 0.05); Thumb, Mid, Ring, Pinky curled (> 0.85)	Tests isolated single-finger pointing
Reem	Small low-amplitude wrist flourish	Index & Middle crossed/flat (f1, f2 ≈ 0.10), Ring/Pinky curled	Validates fine-grained finger separation
Thanks	Forward linear push along Z axis	All 5 fingers strictly flat and rigid (all f < 0.05)	Differentiates wave from flat push
Thank you	Forward outward arc	Starts flat (f ≈ 0.05), curls into partial cup during release	Tests multi-stage flex trajectory
Hello Reem	Compound dual-peak wrist trajectory	Transitions from open wave into "R" finger configuration	Tests compound gesture classification
I am Reem	Compound point then small flourish	Index point (f1=0) transitions into name finger signature	Validates compound multi-stage posture

10. Phase 2 Implementation Roadmap Checklist

Track A: Hardware & Electronics

- Procure 5× 2.2" or 4.5" bi-directional flex sensors
- Build 5× 10 kΩ 1% metal film divider circuit on wrist protoboard
- Solder wiring loom to ESP32 ADC1 pins (34, 35, 32, 33, 36)
- Mount sensors to dorsal glove fabric with flexible adhesive backing
- Validate ADC delta > 1200 counts between flat and 90° bend

Track B: Firmware Engineering

- Write sender_v2.ino with 25 Hz ADC1 polling & EWMA filter
- Implement guided serial/button calibration with NVS storage
- Expand ESP-NOW packet to 33-byte SensorData struct
- Write mqtt_sender_v2.ino with 512-byte buffer & 9-field CSV
- Verify zero packet drops across 10,000 continuous transmissions

Track C: Node-RED Backend Pipeline

- Upgrade serial-manager.js with 9-field parser & v1/v2 detector
- Implement 5-channel synthetic flex patterns in simulator
- Update gesture-engine.js buffer to store 9-element sample rows
- Expand dataset-model.js feature extraction from 66D to 121D
- Deploy flows via automated script node-red/build-flows.js

Track D: Dashboard UI & Validation

- Build 5-finger color-coded real-time bar gauge on Live panel
- Add finger bend timeline chart to Last Gesture trace card
- Implement Calibration Wizard modal with countdown bars
- Update 121D grouped visualizer in Feature Inspector modal
- Collect 20 samples per sign; evaluate leave-one-out accuracy > 92%

11. Risk Matrix & Engineering Mitigations

Identified Technical Risk	Probability	Impact	Architectural Mitigation Strategy
ADC2 Wi-Fi Radio Collision	Certain	Critical	Strictly isolate flex sensors to ADC1 (GPIO 32–36). ADC2 pins are prohibited in firmware.
ESP32 ADC Non-Linearity	High	Medium	On-device two-point calibration normalizes active travel range; divider tuned for 0.3V–2.8V linear zone.
Fabric Creep / Sensor Drift	Medium	Medium	Glove calibration takes 6 seconds and can be re-run on demand; dashboard flags uncalibrated offsets.
Sensor Hardware Failure	Low	High	Dashboard settings provide per-channel software disables; feature extractor imputes zero variance for dead channels.
Higher Dimensionality Overfitting	Medium	Medium	Increase training density from 3 to 15+ samples per word; adjust k-NN neighbors from k=3 to k=5.

📌 12. Summary of Architectural Design Decisions

System Sub-System	Phase 2 Architectural Decision	Engineering Rationale & System Benefit
Gesture Triggering	Gyroscope Magnitude Only	Flex sensors change slowly and are prone to baseline jitter; gyro magnitude provides crisp motion boundaries.
Sampling Cadence	Synchronous 25 Hz (40 ms)	Samples gyro and all 5 flex sensors simultaneously in a single loop, eliminating temporal phase skew.
Calibration Domain	On-Transmitter (Firmware)	Normalizing to 0.0–1.0 on-device saves transmission bandwidth and keeps backend processing clean.
Feature Vector Scale	121 Dimensions (66D + 55D)	Captures distribution (25D), static pose (5D), onset (10D), and temporal coarse trajectory (15D).
Dataset Management	Clean Segregation	Zero-padding Phase 1 recordings biases the 121D Euclidean manifold; models must use dedicated Phase 2 sets.